Supporting document for reviewers:

Exact aniGTO electron-repulsion integrals at extreme anisotropy by the logarithmic variable $s = \ln u$ in the Boys-type integral

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Companion to the manuscript "Probing Dimensional Crossover in SCAN Exchange with Exact Quasi-One-Dimensional Wires and Quasi-Two-Dimensional Sheets" and its Supporting Information

What is new, in four points.

1. *Problem.* aniGTO evaluates every anisotropic two-electron repulsion integral (ERI) as one Boys-type integral over $t = u/\sqrt{1+u^2} \in [0, 1)$ (eq N21). An exponent ratio ρ puts its contribution at $1 - t \approx 1/(2\rho)$: it is under-resolved from $\rho \sim 10^8$, and beyond $u^2 = 2^{52} \approx 4.5 \times 10^{15}$ no double-precision $t < 1$ exists at all. The default routine then fails by up to 11.3 Ha (Table N1).
2. *What we solved.* The same integral is rewritten in the scale-free variable $s = \ln u$ (eq N24 = main-text eq 1). The closed-form aniGTO kernels I_k (eq N18, derived step by step in Section 2) are unchanged; we add explicit truncation bounds and an analytic small-u piece (Section 4).
3. *Proof of accuracy.* On the exactly solvable self-repulsion of anisotropic Gaussians (closed forms, main eqs A10 and A11) the error is $< 10^{-11}$ mHa, the double-precision limit, for wire and sheet at every $\lambda = 10^1$–10^8, i.e. exponent ratios up to 10^{16} (Table N1, Figure N1).
4. *Hamiltonian and code.* The orbitals are exact eigenfunctions of the one-particle trap Hamiltonian, solved analytically; no Kohn–Sham equation is solved (Section 1). All code is in `SCAN_1D_2D/codes/01-09`; the ERI is in `04_exact_vs_proposed.py`, listed in Codes 2–5 (Section 8).

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1 The one-particle Hamiltonian: where the orbitals come from

No Kohn–Sham (or Hartree–Fock) equation is solved self-consistently anywhere in this work, and no script calls an SCF solver. All results are *fixed-density, exchange-only*: exchange functionals are evaluated on densities whose orbitals are known exactly. Those orbitals are the eigenfunctions of the one-particle anisotropic harmonic-trap Hamiltonian

$$\hat{h} = -\tfrac{1}{2}\nabla^2 + \tfrac{1}{2}\sum_{k=x,y,z} \omega_k^2 x_k^2, \tag{N1}$$

which separates into three one-dimensional oscillators and is solved *analytically*:

$$\hat{h}\,\phi_{\mathbf{n}} = \varepsilon_{\mathbf{n}}\phi_{\mathbf{n}}, \qquad \phi_{\mathbf{n}}(\mathbf{r}) = \prod_k \omega_k^{1/4}\, h_{n_k}\big(\sqrt{\omega_k}\, x_k\big), \qquad \varepsilon_{\mathbf{n}} = \sum_k \omega_k\big(n_k + \tfrac{1}{2}\big), \tag{N2}$$

with the Hermite functions h_n of SI eq S4. The ground state is the normalized aniGTO s function of the harmonic-trap ("dot") basis, with exponents $\alpha_k = \omega_k/2$ (SI eqs S1–S3):

$$\phi_{\mathbf{0}}(\mathbf{r}) = \prod_k \Big(\frac{\omega_k}{\pi}\Big)^{1/4} e^{-\omega_k x_k^2/2}, \qquad n(\mathbf{r}) = 2\phi_{\mathbf{0}}^2, \tag{N3}$$

which is the reference density of the manuscript: $\boldsymbol{\omega} = (1, 1, \lambda^2)$ for the sheet and $(\lambda^2, \lambda^2, 1)$ for the wire (main eqs A7 and A8). For $N = 2$ (one doubly occupied orbital) the exact exchange energy is a single ERI (main eq A9),

$$E_x = -\tfrac{1}{2}E_H = -(\phi_{\mathbf{0}}\phi_{\mathbf{0}}|\phi_{\mathbf{0}}\phi_{\mathbf{0}}), \tag{N4}$$

which is why the accuracy of that one integral decides the benchmark. The $N = 4$ and 6 wires of the SI occupy the s, p_z, d_{zz} eigenfunctions ($n_z = 0, 1, 2$) of the same $\hat{h}$. Because $\hat{h}$ is solved in closed form there is no basis-set or SCF error in any reference. The self-consistent potential of DC-SCAN is derived in main Appendix A.9 but is not used here: self-consistent runs are future work.

Code 1 shows how script `03` builds this orbital (class `TrapS`, with exponents $\omega_k/2$ and normalization $\prod_k (2\alpha_k/\pi)^{1/4}$), takes the exact exchange from the aniGTO ERI (`build_wire`), and forms n, ∇n, $\nabla\nabla n$ and τ from the analytic derivatives (`density_fields`; SI eqs S16 and S17). PySCF enters only through `eval_xc`, which evaluates libxc exchange functionals on these fields.

Code 1: The analytic trap orbital and its density fields (script 03, verbatim).

```python
# codes/03_anigto_wire_libxc.py (verbatim)

class TrapS:
    """The s function of the aniGTO harmonic-trap ('dot') basis for omega = (wx, wy, wz): a normalized anisotropic
    Gaussian with exponents omega/2, i.e. phi(r) = prod_k (omega_k/pi)^(1/4) exp(-omega_k x_k^2 / 2).
    Attributes follow the aniGTO BasisFunction object (exps, shell, origin, coefs, norm, num_coefs)."""

    def __init__(self, omega):
        a = 0.5 * np.asarray(omega, dtype=float)
        self.exps = a.reshape(1, 3)
        self.shell = (0, 0, 0)
        self.origin = (0.0, 0.0, 0.0)
        self.coefs = np.array([1.0])
        self.norm = np.array([np.prod((2.0 * a / np.pi) ** 0.25)])
        self.num_coefs = 1

class TrapMolecule:
    """Minimal stand-in for aniGTO Molecule(basis='dot', n_orbitals={'S': 1}): one trap s function. """

    def __init__(self, omega):
        self.bfs = [TrapS(omega)]

def build_wire(lam, workdir=None):
    """Wire trap omega = (lam^2, lam^2, 1); returns (molecule, exact E_x = -(phi phi|phi phi) from aniGTO IEE)."""
    mol = TrapMolecule([lam ** 2, lam ** 2, 1.0])
    b = mol.bfs[0]
    a, L, O = tuple(b.exps[0]), (0, 0, 0), (0.0, 0.0, 0.0)
    with warnings.catch_warnings():
        warnings.simplefilter('ignore')
        eri = b.norm[0] ** 4 * IEE(a, L, O, a, L, O, a, L, O, a, L, O)
    assert np.allclose(b.exps[0], [lam ** 2 / 2, lam ** 2 / 2, 0.5]), b.exps
    return mol, -eri

def density_fields(mol, coords):
    """rho, grad rho, Hessian of rho, laplacian, tau for the doubly occupied aniGTO s orbital."""
    phi, g, h = eval_bf_hess(mol.bfs[0], coords)
    rho = 2.0 * phi * phi
    grad = 4.0 * phi * g
    hess = 4.0 * (g[:, None, :] * g[None, :, :] + phi * h)
    lap = np.trace(hess)
    tau = np.sum(g * g, axis=0)                          # 1/2 sum_occ f |grad phi|^2 with f = 2
    return rho, grad, hess, lap, tau
```

2 Derivation of the aniGTO repulsion integral

The derivation follows the aniGTO notes (SI Section S1) and uses only three tools. Each step names the code line that implements it (Code 2).

2.1 Tools and notation

Gaussian product theorem.

$$e^{-\alpha(x-A)^2}e^{-\beta(x-B)^2} = E_{AB}\,e^{-(\alpha+\beta)(x-P)^2}, \qquad P = \frac{\alpha A+\beta B}{\alpha+\beta}, \qquad E_{AB} = e^{-\frac{\alpha\beta}{\alpha+\beta}(A-B)^2}. \tag{N5}$$

Binomial theorem.

$$(x+a)^n = \sum_{i=0}^{n}\binom{n}{i}x^i a^{n-i}, \tag{N6}$$

applied factor by factor to products $(x+a)^n(x+b)^m\cdots$.

Gaussian moments (`integral(a,b)` with `fact2`):

$$\int_{-\infty}^{+\infty} x^n e^{-\alpha x^2}dx = \sqrt{\frac{\pi}{\alpha}}\,\frac{(n-1)!!}{(2\alpha)^{n/2}} \quad (n \text{ even}), \qquad 0 \quad (n \text{ odd}), \qquad (-1)!! = 1. \tag{N7}$$

Combining eqs N6 and N7, a product of four shifted monomials under one Gaussian centred at U is (`integral2(alpha,U,a,A,b,B,c,C,d,D)`)

$$\int\prod_{i=1}^{4}(x-X_i)^{n_i}e^{-\beta(x-U)^2}dx = \sum_{j_1\ldots j_4}\prod_{i=1}^{4}\binom{n_i}{j_i}(U-X_i)^{n_i-j_i}\sqrt{\frac{\pi}{\beta}}\frac{(J-1)!!}{(2\beta)^{J/2}}, \quad J=\textstyle\sum_i j_i \text{ even}. \tag{N8}$$

Anisotropic Gaussians. A primitive has one exponent and one power per axis,

$$G_A(\mathbf{r},\boldsymbol{\alpha},\mathbf{n},\mathbf{A}) = \prod_{k=x,y,z}(x_k-A_k)^{n_k}e^{-\alpha_k(x_k-A_k)^2}. \tag{N9}$$

2.2 The two-electron repulsion integral and its factorization

$$I^{EE}_{ABCD} = \iint d\mathbf{r}_1 d\mathbf{r}_2\,\frac{G_A(\mathbf{r}_1,\boldsymbol{\alpha},\mathbf{n},\mathbf{A})\,G_B(\mathbf{r}_1,\boldsymbol{\beta},\mathbf{m},\mathbf{B})\,G_C(\mathbf{r}_2,\boldsymbol{\gamma},\mathbf{p},\mathbf{C})\,G_D(\mathbf{r}_2,\boldsymbol{\delta},\mathbf{q},\mathbf{D})}{r_{12}}. \tag{N10}$$

The Laplace (Gaussian) transform of the Coulomb operator (SI eq S8),

$$\frac{1}{r_{12}} = \frac{2}{\sqrt{\pi}}\int_0^\infty e^{-u^2 r_{12}^2}\,du, \qquad r_{12}^2 = \sum_k (x_{1k}-x_{2k})^2, \tag{N11}$$

turns $1/r_{12}$ into a product over axes. With eq N5 for the pairs (A,B) on electron 1 and (C,D) on electron 2 the six-dimensional integral factorizes for every u:

$$I^{EE}_{ABCD} = \frac{2}{\sqrt{\pi}}\,E\int_0^\infty du\,I_x(u^2)\,I_y(u^2)\,I_z(u^2), \qquad E = \prod_k E_{k,AB}E_{k,CD}, \tag{N12}$$

$$I_x(u^2) = \iint dx_1 dx_2\,(x_1-A)^n(x_1-B)^m(x_2-C)^p(x_2-D)^q \times e^{-(\alpha+\beta)(x_1-P_{AB})^2}e^{-(\gamma+\delta)(x_2-P_{CD})^2}e^{-u^2(x_1-x_2)^2}, \tag{N13}$$

where the axis label is dropped on $\alpha,\ldots,q$ and $A,\ldots,D$ (SI eqs S9 and S10). This is the structure of `IEE`: the prefactor E (`ExAB…EzCD`) times the u integral of the product of three calls of `integral3`.

2.3 The x_1 integral

Completing the square in x_1,

$$e^{-(\alpha+\beta)(x_1-P_{AB})^2-u^2(x_1-x_2)^2} = e^{-(\alpha+\beta+u^2)(x_1-Q)^2}\, e^{-\xi(x_2-P_{AB})^2},$$
$$Q = \frac{(\alpha+\beta)P_{AB}+u^2x_2}{\alpha+\beta+u^2}, \qquad \xi = \frac{(\alpha+\beta)u^2}{\alpha+\beta+u^2} \tag{N14}$$

(`xi`, `PAB` in `integral3`). Writing $x_1 - A = (x_1 - Q) + (Q - A)$, expanding binomially and using eq N7 for $x_1 - Q$:

$$\int dx_1 (x_1-A)^n (x_1-B)^m e^{-(\alpha+\beta+u^2)(x_1-Q)^2}$$
$$= \sqrt{\frac{\pi}{\alpha+\beta+u^2}} \sum_{i,j}^{i+j\ \text{even}} \binom{n}{i}\binom{m}{j} \frac{(i+j-1)!!}{[2(\alpha+\beta+u^2)]^{(i+j)/2}} (Q-A)^{n-i}(Q-B)^{m-j}. \tag{N15}$$

Q depends on x_2; the shifted-centre identity

$$Q - A = \frac{u^2}{\alpha+\beta+u^2}(x_2 - A'), \qquad A'(u^2) = \frac{A(\alpha+\beta+u^2)-(\alpha+\beta)P_{AB}}{u^2} \tag{N16}$$

(and likewise B'; `A1`, `B1` in `integral3`) makes the remaining x_2 integrand a polynomial in x_2 times Gaussians.

2.4 The x_2 integral and the closed form

The two x_2 Gaussians combine by eq N5:

$$e^{-(\gamma+\delta)(x_2-P_{CD})^2} e^{-\xi(x_2-P_{AB})^2} = e^{-\frac{(\gamma+\delta)\xi}{\gamma+\delta+\xi}(P_{CD}-P_{AB})^2}\, e^{-(\gamma+\delta+\xi)(x_2-U)^2},$$
$$U = \frac{(\gamma+\delta)P_{CD}+\xi P_{AB}}{\gamma+\delta+\xi} \tag{N17}$$

(`U` and the exponential in `integral3`). The x_2 integral is then eq N8 with $\beta = \gamma+\delta+\xi$ and the four factors $(x_2-C)^p(x_2-D)^q(x_2-A')^{n-i}(x_2-B')^{m-j}$ (`integral2(c+d+xi,U,n-i,A1,m-j,B1,p,C,q,D)`). Collecting,

$$I_x(u^2) = \sqrt{\frac{\pi}{\alpha+\beta+u^2}}\sqrt{\frac{\pi}{\gamma+\delta+\xi}}\, e^{-\frac{(\gamma+\delta)\xi}{\gamma+\delta+\xi}(P_{CD}-P_{AB})^2}$$
$$\times \sum_{i=0}^{n}\sum_{j=0}^{m} \binom{n}{i}\binom{m}{j} \frac{(i+j-1)!!}{[2(\alpha+\beta+u^2)]^{(i+j)/2}} \left(\frac{u^2}{\alpha+\beta+u^2}\right)^{n+m-i-j}$$
$$\times \sum_{k=0}^{p}\sum_{l=0}^{q}\sum_{r=0}^{n-i}\sum_{s'=0}^{m-j} \binom{p}{k}\binom{q}{l}\binom{n-i}{r}\binom{m-j}{s'} \frac{(k+l+r+s'-1)!!}{[2(\gamma+\delta+\xi)]^{(k+l+r+s')/2}}$$
$$\times (U-C)^{p-k}(U-D)^{q-l}(U-A')^{n-i-r}(U-B')^{m-j-s'}, \tag{N18}$$

with only even $i+j$ and even $k+l+r+s'$ contributing (SI eq S12). *Eq N18 is exact and closed-form in u^2*: no approximation has been made so far, and the only numerical step left is the single u integral of eq N12.

2.5 Connection to the Boys function, and why anisotropy breaks it

For an isotropic s-type quartet ($n = m = p = q = 0$) with $p_1 = \alpha+\beta$, $p_2 = \gamma+\delta$ on every axis, reduced exponent $\varrho = p_1p_2/(p_1 + p_2)$ and $\mathbf{R} = \mathbf{P}_{AB} - \mathbf{P}_{CD}$, eq N18 has one term and $(p_1 + u^2)(p_2 + \xi) = (p_1 + p_2)(\varrho + u^2)$, so

$$\prod_k I_k(u^2) = \frac{\pi^3}{(p_1 + p_2)^{3/2}(\varrho + u^2)^{3/2}} \exp\Big(-\frac{\varrho u^2}{\varrho + u^2} R^2\Big). \tag{N19}$$

The Boys substitution $t_\varrho = u/\sqrt{\varrho + u^2}$, $dt_\varrho = \varrho(\varrho + u^2)^{-3/2}du$, removes the scale exactly:

$$\int_0^\infty \prod_k I_k \, du = \frac{\pi^3}{(p_1 + p_2)^{3/2}\varrho} \int_0^1 e^{-Tt^2} dt = \frac{\pi^3}{(p_1 + p_2)^{3/2}\varrho} F_0(T), \qquad T = \varrho R^2, \tag{N20}$$

which with eq N12 is the familiar $(ss|ss) = 2\pi^{5/2} E\, F_0(T)/(p_1p_2\sqrt{p_1 + p_2})$ of Boys [1]; the integrand e^{-Tt^2} is smooth on $[0, 1]$ for any exponents.

For anisotropic primitives every axis has its own reduced exponent ϱ_k, and $\prod_k I_k \propto \prod_k(\varrho_k + u^2)^{-1/2} \times (\ldots)$. *No single substitution removes three different scales.* aniGTO therefore uses the fixed scale $\varrho = 1$, $t = u/\sqrt{1 + u^2}$, $u^2 = t^2/(1 - t^2)$, $du = (1 - t^2)^{-3/2}dt$ (SI eq S13):

$$I^{EE}_{ABCD} = \frac{2}{\sqrt{\pi}} E \int_0^1 \frac{dt}{(1 - t^2)^{3/2}} I_x\Big(\frac{t^2}{1 - t^2}\Big) I_y\Big(\frac{t^2}{1 - t^2}\Big) I_z\Big(\frac{t^2}{1 - t^2}\Big), \tag{N21}$$

evaluated by adaptive quadrature (`scipy.integrate.quad`, tolerances 10^{-16}; `integrand2` and `IEE`, Code 3). This is the "aniGTO default" baseline of Table N1.

3 Why the default t quadrature fails at large anisotropy

(i) Location. Axis k changes the integrand where $u^2 \sim \varrho_k$. Since $1 - t = 1 - u/\sqrt{1 + u^2} \simeq 1/(2u^2)$ for $u \gg 1$, an exponent scale ϱ is mapped to

$$1 - t \simeq \frac{1}{2\varrho}. \tag{N22}$$

Exponent ratios 10^8 and 10^{16} (wire $\lambda = 10^4$ and 10^8) put the feature at $1 - t \sim 10^{-8}$ and 10^{-16}.

(ii) Shape: a near-singularity in t. For the wire self-repulsion ($A = 2\boldsymbol{\alpha} = \mathrm{diag}(\lambda^2, \lambda^2, 1)$, eq N27 below) the u integrand is $g(u) = \lambda^2/[(\lambda^2 + 2u^2)\sqrt{1 + 2u^2}] \simeq 1/(\sqrt{2}\, u)$ for $1 \ll u \ll \lambda$. With $1 - t = \delta$, $u \simeq (2\delta)^{-1/2}$ and $du/dt = (1 - t^2)^{-3/2} \simeq (2\delta)^{-3/2}$, the t integrand is

$$g\,\frac{du}{dt} \simeq \frac{1}{2\sqrt{2}\,(1 - t)}, \qquad \frac{1}{\lambda^2} \lesssim 1 - t \ll 1, \tag{N23}$$

a $1/(1 - t)$ near-singularity cut off only at $1 - t \sim \lambda^{-2}$. It carries the whole wire logarithm, $\int d\delta/(2\sqrt{2}\,\delta) = \ln\lambda/\sqrt{2}$, which after the prefactor $2/\sqrt{\pi}$ is the exact $\sqrt{2/\pi}\ln\lambda$ of main eq A13. An adaptive rule must resolve $\sim \ln\lambda^2$ decades of $1 - t$ next to the endpoint.

(iii) Round-off and the end of double precision. The kernel argument $u^2 = t^2/(1 - t^2)$ is computed from the difference $1 - t^2$, whose relative round-off is $\epsilon/(1 - t^2) \approx 2\epsilon u^2$ ($\epsilon = 2^{-53}$): $\approx 2 \times 10^{-6}$ relative error in u^2 at $u^2 = 10^{10}$. The largest double below 1 is $t_{\max} = 1 - 2^{-53}$, for which $u^2_{\max} = t^2_{\max}/(1 - t^2_{\max}) = 2^{52} - 1 \approx 4.5 \times 10^{15}$: larger u cannot be represented in t at all, while the wire at $\lambda = 10^8$ needs u^2 up to $\sim \lambda^2/2 = 5 \times 10^{15}$.

Observed consequence (Table N1). The default is exact to 3×10^{-9} mHa up to $\lambda = 10^3$, loses 0.067 mHa (wire) and 0.080 mHa (sheet) at $\lambda = 10^4$, and for $\lambda \geq 10^5$ the wire value freezes at -3.906 Ha (errors 5833 to 11344 mHa). On the sheet the default returns $-1.253314137 = -\sqrt{\pi/2}$, the $\lambda \to \infty$ limit, for every $\lambda \geq 10^5$: it misses exactly the $\sqrt{2/\pi}\,\lambda^{-1}$ term that comes from $u \sim \lambda$ (error 8.0×10^{-3} mHa at 10^5, decreasing as λ^{-1}). The short wall times of the default at $\lambda \geq 10^5$ (5 ms) are this failure, not a speed-up.

4 The proposed method: the Boys-type integral in $s = \ln u$

4.1 The equation that is solved

Because $dt\,(1-t^2)^{-3/2} = du = e^s\,ds$, eq N21 is identically

$$I^{EE}_{ABCD} = \frac{2}{\sqrt{\pi}}\,E \int_{-\infty}^{\infty} f(s)\,ds, \qquad f(s) = e^s\,I_x\big(e^{2s}\big)\,I_y\big(e^{2s}\big)\,I_z\big(e^{2s}\big)\,, \qquad s = \ln u, \tag{N24}$$

which is main-text eq 1. The aniGTO kernels are unchanged: I_k is still the closed form eq N18 (`integral3`), now called with $u^2 = e^{2s}$, computed without any cancellation.

4.2 Why s is the right variable

s is scale-free: changing an exponent scale $\varrho_k \to c\,\varrho_k$ shifts the feature of axis k by $\frac{1}{2}\ln c$ in s without changing its shape. Every axis therefore produces an $O(1)$-wide feature at $s \approx \frac{1}{2}\ln \varrho_k$, whatever the anisotropy. The near-singularity eq N23 becomes a flat plateau: for the wire, $f(s) = u\,g(u) \simeq 1/\sqrt{2}$ for $0 \ll s \ll \ln\lambda$, a constant over $\ln\lambda$ units of s that any quadrature integrates exactly. The representability limit (iii) disappears: e^{2s} is representable up to $u^2 \sim 10^{308}$.

4.3 Finite range, end pieces and error bounds

With $\alpha_{\min}, \alpha_{\max}$ the smallest and largest of the twelve primitive exponents,

$$s \in [s_0, s_1], \qquad s_0 = \tfrac{1}{2}\ln\alpha_{\min} - 20, \qquad s_1 = \tfrac{1}{2}\ln\alpha_{\max} + 20. \tag{N25}$$

Small-u piece $[0, u_0]$, $u_0 = e^{s_0}$: $\prod_k I_k(u^2)$ is analytic in u^2, so $\int_0^{u_0} \prod_k I_k\,du = u_0 \prod_k I_k(u_0^2)\,[1 + O(u_0^2/\alpha_{\min})] = f(s_0)\,[1 + O(e^{-40})]$, added analytically. *Large-u tail* $[e^{s_1}, \infty)$: for the s-type integrals tested here each $I_k \propto u^{-1}$ at large u, so the neglected tail is $O(\alpha_{\max} e^{-2s_1}) = O(e^{-40})$ relative. Both truncations are thus $\sim e^{-40} \approx 4 \times 10^{-18}$ relative, below double precision.

4.4 Quadrature

The integral over $[s_0, s_1]$ uses `scipy.integrate.quad` with breakpoints at every integer s (unit spacing, so each $O(1)$ feature is its own panel), `limit=4000`, `epsrel=1e-13`, `epsabs=0`:

$$I^{EE}_{ABCD} = \frac{2}{\sqrt{\pi}}\,E\Big[\int_{s_0}^{s_1} f(s)\,ds + f(s_0)\Big]. \tag{N26}$$

This is `IEE_robust` (Code 4), a drop-in replacement for aniGTO's `IEE` with the same arguments.

5 Exact test and results

5.1 The exactly solvable test integral

For four identical s functions (eq N3) eq N18 reduces to $I_k = \pi/[2\sqrt{\alpha_k(\alpha_k+u^2)}]$ (SI eq S14) and eq N12 to

$$(\phi\phi|\phi\phi) = \frac{2}{\sqrt{\pi}}\int_0^\infty \frac{\sqrt{\det A}}{\sqrt{\det(A+2u^2 I)}}\,du, \qquad A = 2\,\mathrm{diag}(\boldsymbol{\alpha}) = \mathrm{diag}(\boldsymbol{\omega}), \tag{N27}$$

(SI eq S15 = main eq A44), whose closed forms are main eqs A10 (sheet) and A11 (wire). The test (`self_repulsion`, Code 5) runs the *general* aniGTO code path, `integral3` for all three axes, and compares with these closed forms.

5.2 Results

Table N1 (blocks a, b) and Figure N1(a, b) are the main-text Table 4 and Figure 1, generated by `04_exact_vs_proposed.py`. The proposed method reproduces the closed form to $< 10^{-11}$ mHa at every λ for both families, including the wire at $\lambda = 10^8$ (exact -15.250628199 Ha), where the default returns -3.906 Ha. On an ordinary anisotropic same-centre quartet with Cartesian powers $(2,0,1)$ and $(0,1,1)$ (exponent ratio 5) the two routes agree to machine precision (difference 0.0 in `results/iee_robust.csv`) at 283 versus 495 ms, about 1.7 times the cost; for the test integrals the cost is 70–99 ms per integral. Block (c) and panel (c) concern the $N = 2$ strictly-correlated solver of the manuscript and are shown for completeness. All tests use quartets on one centre (the trap origin); multi-centre integrals use the same unchanged kernel eq N18.

Table N1: Exact references versus the proposed methods (main-text Table 4, generated by `codes/04_exact_vs_proposed.py`). (a, b) $E_x = -(\phi\phi|\phi\phi)$ of the two-electron Gaussian wire and sheet (hartree): exact closed form, aniGTO default quadrature in t (eq N21, baseline) and $s = \ln u$ quadrature (eq N26, proposed). Errors in mHa; $< 10^{-11}$ mHa is the double-precision limit. Time: wall time of one integral.

			baseline	proposed	$\lvert E - E^{\mathrm{exact}}\rvert$ (mHa)		time per integral (ms)	
quantity	parameter	exact	(see block)	(see block)	of baseline	of proposed	of baseline	of proposed
(a) wire, $E_x = -(\phi\phi\|\phi\phi)$. Exact: eq A11; baseline: aniGTO default (eq S13, adaptive in t); proposed: aniGTO kernels in $s = \ln u$ (eq 1)								
wire	$\lambda = 10^1$	-2.400277840	-2.400277840	-2.400277840	$< 10^{-11}$	$< 10^{-11}$	20	73
wire	$\lambda = 10^2$	-4.227637064	-4.227637064	-4.227637064	3.2×10^{-11}	$< 10^{-11}$	40	79
wire	$\lambda = 10^3$	-6.064645553	-6.064645553	-6.064645553	3.0×10^{-9}	$< 10^{-11}$	55	80
wire	$\lambda = 10^4$	-7.901839854	-7.901906791	-7.901839854	0.067	$< 10^{-11}$	68	81
wire	$\lambda = 10^5$	-9.739036912	-3.906470773	-9.739036912	5833	$< 10^{-11}$	5	88
wire	$\lambda = 10^6$	-11.576234008	-3.906470882	-11.576234008	7670	$< 10^{-11}$	5	93
wire	$\lambda = 10^7$	-13.413431103	-3.906470883	-13.413431103	9507	$< 10^{-11}$	5	94
wire	$\lambda = 10^8$	-15.250628199	-3.906470883	-15.250628199	11344	$< 10^{-11}$	5	99
(b) sheet, $E_x = -(\phi\phi\|\phi\phi)$. Exact: eq A10; baseline and proposed as in (a)								
sheet	$\lambda = 10^1$	-1.179303431	-1.179303431	-1.179303431	$< 10^{-11}$	$< 10^{-11}$	54	70
sheet	$\lambda = 10^2$	-1.245397430	-1.245397430	-1.245397430	$< 10^{-11}$	$< 10^{-11}$	41	74
sheet	$\lambda = 10^3$	-1.252516879	-1.252516879	-1.252516879	$< 10^{-11}$	$< 10^{-11}$	62	79
sheet	$\lambda = 10^4$	-1.253234355	-1.253314143	-1.253234355	0.080	$< 10^{-11}$	61	83
sheet	$\lambda = 10^5$	-1.253306159	-1.253314137	-1.253306159	8.0×10^{-3}	$< 10^{-11}$	30	88
sheet	$\lambda = 10^6$	-1.253313339	-1.253314137	-1.253313339	8.0×10^{-4}	$< 10^{-11}$	21	94
sheet	$\lambda = 10^7$	-1.253314058	-1.253314137	-1.253314058	8.0×10^{-5}	$< 10^{-11}$	18	92
sheet	$\lambda = 10^8$	-1.253314129	-1.253314137	-1.253314129	7.9×10^{-6}	$< 10^{-11}$	17	97
(c) $N = 2$ SCE, V_{ee}^{SCE}. Exact: eq 4; baseline: optimal transport (eq 2), finest grid; proposed: $h \to 0$, eq 3								
Gaussian	$\omega = 1$	0.444142598	0.444522674	0.444172297	0.380	0.030	–	–
Gaussian	$\omega = 0.1$	0.140450222	0.140570412	0.140459613	0.120	9.4×10^{-3}	–	–

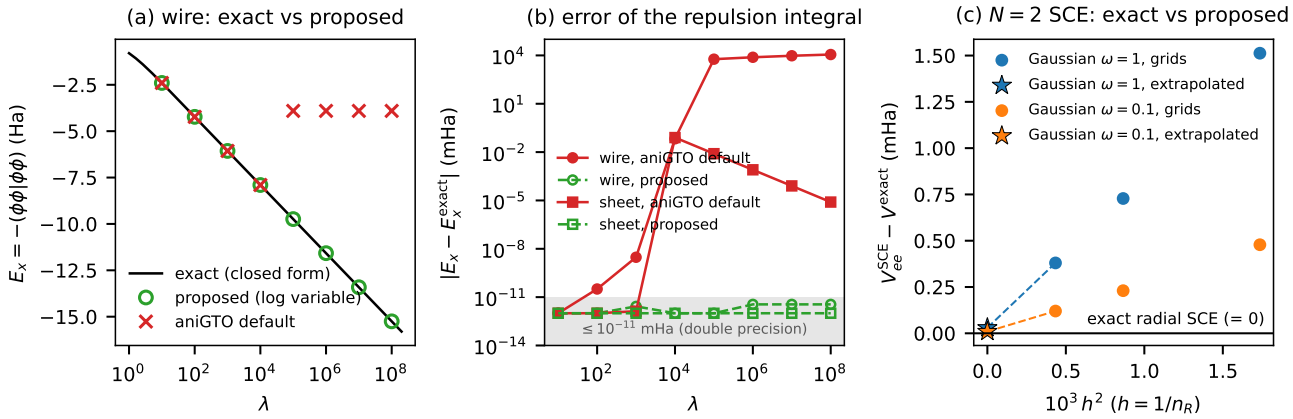


Figure N1: (a) Wire self-repulsion versus λ: closed form (line), $s = \ln u$ quadrature (circles) and aniGTO default (crosses), which collapses to -3.906 Ha for $\lambda \geq 10^5$. (b) Absolute errors (mHa) for wire and sheet; the shaded band is the double-precision limit. (c) $N = 2$ SCE check of the manuscript (not part of the ERI method). Main-text Figure 1.

6 The code, verbatim

The listings below are copied automatically from `codes/04_exact_vs_proposed.py` at every build (`manuscript/make_eri_snippets.py`), so they are the code that produced Table N1. The first two are copied verbatim from aniGTO; only `IEE_robust` is new.

Code 2: aniGTO kernel: Gaussian moments (eq N7), products of four monomials (eq N8) and the closed-form $I_k(u^2)$ (eqs N14–N18).

```
# codes/04_exact_vs_proposed.py (verbatim)

def fact2(n):
    """Calculate the double factorial (n - 1)!! of a non-negative integer 'n'."""
    result = 1
    for i in range(n, 1, -2):
        result *= i
    return result

def integral(a,b):
    r"""this function evaluate an integral like this:
    int_{-\infty}^{+\infty}dx x^a e^{-bx^2}
    """
    if a%2==0.00000000000000000:
        return np.sqrt(np.pi/b)*fact2(a-1)/(2*b)**(a/2)
    else:
        return 0.000000000000000000000

def integral2(alpha,U,a,A,b,B,c,C,d,D):
    """this function evaluates an integral like the one above"""
    N=0.000000000000000000000
    for i in range(a+1):
        for j in range(b+1):
            for k in range(c+1):
                for r in range(d+1):
                    N=N+comb(a,i)*comb(b,j)*comb(c,k)*comb(d,r)*(U-A)**(a-i)*(U-B)**(b-j)*(U-C)**(c
    ↪ -k)*(U-D)**(d-r)*integral(i+j+k+r,alpha)
```

```
    return N

def integral3(u2,a,n,A,b,m,B,c,p,C,d,q,D):
    """evaluates the integral over x_1 and x_2 only
    which is the on above"""
    xi=(a+b)*u2/(a+b+u2)
    PAB=(a*A+b*B)/(a+b)
    PCD=(c*C+d*D)/(c+d)
    U=((c+d)*PCD+xi*PAB)/(c+d+xi)
    A1=(A*(a+b+u2)-(a+b)*PAB)/u2
    B1=(B*(a+b+u2)-(a+b)*PAB)/u2
    N=0.00000000000000000000000000000
    for i in range(n+1):
        for j in range(m+1):
            if (i+j)%2==0:
                N=N+comb(n,i)*comb(m,j)*fact2(i+j-1)/(2*a+2*b+2*u2)**((i+j)/2)*((u2)/(a+b+u2))**(n+
    ↪ m-i-j)*np.exp(-(c+d)*xi/(c+d+xi)*(PAB-PCD)**2)*integral2(c+d+xi,U,n-i,A1,m-j,B1,p,C,q,D)
            else:
                N=N+0.00000000000000000000000000000
    return np.sqrt(np.pi/(a+b+u2))*N
```

Code 3: aniGTO default: the Boys-type integral in t (eq N21); baseline of Table N1.

```
# codes/04_exact_vs_proposed.py (verbatim)

def integrand2(t, a, l1, A, b, l2, B, c, l3, C, d, l4, D):
    """Define the integrand for numerical integration."""
    ax, ay, az = a
    nx, ny, nz = l1
    Ax, Ay, Az = A
    bx, by, bz = b
    mx, my, mz = l2
    Bx, By, Bz = B
    cx, cy, cz = c
    px, py, pz = l3
    Cx, Cy, Cz = C
    dx, dy, dz = d
    qx, qy, qz = l4
    Dx, Dy, Dz = D
    u2 = t**2 / (1 - t**2)
    return (
        np.power(1 - t**2, -1.5000000000000000000000000000000000000000)
        * integral3(u2, ax, nx, Ax, bx, mx, Bx, cx, px, Cx, dx, qx, Dx)
        * integral3(u2, ay, ny, Ay, by, my, By, cy, py, Cy, dy, qy, Dy)
        * integral3(u2, az, nz, Az, bz, mz, Bz, cz, pz, Cz, dz, qz, Dz)
    )

def IEE(a,l1,A,b,l2,B,c,l3,C,d,l4,D):
    ax,ay,az=a
    Ax,Ay,Az=A
    bx,by,bz=b
    Bx,By,Bz=B
    cx,cy,cz=c
    Cx,Cy,Cz=C
    dx,dy,dz=d
    Dx,Dy,Dz,=D
    ExAB=np.exp(-ax*bx*(Ax-Bx)**2/(ax+bx))
    EyAB=np.exp(-ay*by*(Ay-By)**2/(ay+by))
    EzAB=np.exp(-az*bz*(Az-Bz)**2/(az+bz))
    ExCD=np.exp(-cx*dx*(Cx-Dx)**2/(cx+dx))
```

```
    EyCD=np.exp(-cy*dy*(Cy-Dy)**2/(cy+dy))
    EzCD=np.exp(-cz*dz*(Cz-Dz)**2/(cz+dz))

    integral,error=quad(integrand2,0.0000000000000,1.0000000000000000000000,(a,l1,A,b,l2,B,c,l3,C,
    ↪ d,l4,D),epsrel =1e-16, epsabs= 1e-16)


    return 2.0000000000000000000000/np.sqrt(np.pi)*ExAB*EyAB*EzAB*ExCD*EyCD*EzCD*integral
```

Code 4: **Proposed**: the same integral in $s = \ln u$ with range, breakpoints and the analytic small-u piece (eqs N24–N26).

```
# codes/04_exact_vs_proposed.py (verbatim)

def IEE_robust(a, l1, A, b, l2, B, c, l3, C, dd, l4, D):
    """Drop-in replacement for IEE.  Same aniGTO 1-D kernel (integral3), but the Boys-type
    integral is taken in s = ln u instead of t = u / sqrt(1 + u^2):  dt (1 - t^2)^(-3/2) = du, so
        int_0^1 integrand2(t) dt = int ds e^s prod_k integral3(e^{2s}, ...)_k.
    In t the region u^2 ~ exponent ratio sits at 1 - t ~ 1/(2u^2) and is lost to round-off (and 1 -
    ↪  t^2
    underflows to 0) once the ratio exceeds ~1e10; in s it is an ordinary O(1)-wide feature.
    Range: s in [ln sqrt(min exp) - 20, ln sqrt(max exp) + 20], unit breakpoints, plus u0 f(0) for
    ↪ [0, u0]."""
    E = 1.0
    for (p, P), (q, Q) in (((a, A), (b, B)), ((c, C), (dd, D))):
        for k in range(3):
            E *= np.exp(-p[k] * q[k] * (P[k] - Q[k]) ** 2 / (p[k] + q[k]))

    def f(s):
        u2 = np.exp(2.0 * s)
        return np.exp(s) * np.prod([integral3(u2, a[k], l1[k], A[k], b[k], l2[k], B[k], c[k], l3[k
    ↪ ], C[k],
                                              dd[k], l4[k], D[k]) for k in range(3)])
    ex = np.array([a, b, c, dd], dtype=float)
    lo, hi = 0.5 * np.log(ex.min()) - 20.0, 0.5 * np.log(ex.max()) + 20.0
    val, _ = quad(f, lo, hi, points=np.arange(np.floor(lo) + 1.0, hi, 1.0), limit=4000, epsrel=1e
    ↪ -13, epsabs=0.0)
    return 2.0 / np.sqrt(np.pi) * E * (val + f(lo))
```

Code 5: Exact test: $E_x = -(\phi\phi|\phi\phi)$ of the wire and sheet (eqs N4 and N27); `fn` is `IEE` or `IEE_robust`.

```
# codes/04_exact_vs_proposed.py (verbatim)

def diag_A(family, lam):
    return (lam ** 2, lam ** 2, 1.0) if family == 'wire' else (1.0, 1.0, lam ** 2)

def self_repulsion(family, lam, fn):
    Adiag = np.array(diag_A(family, lam))
    a = tuple(Adiag / 2.0)
    O, L = (0.0, 0.0, 0.0), (0, 0, 0)
    norm4 = np.prod(Adiag) / np.pi ** 3
    return -norm4 * fn(a, L, O, a, L, O, a, L, O, a, L, O)
```

Usage, e.g. the wire at $\lambda = 10^8$ (exponent ratio 10^{16}):

```
ex = self_repulsion('wire', 1e8, IEE_robust)   # -15.250628199 Ha, closed form eq A11
ex = self_repulsion('wire', 1e8, IEE)          # -3.906470883 Ha: the default fails
```

7 Equation map: what was solved and where it is in the code

Table N2: Equations of this note, their place in the manuscript/SI, and the function that implements them.

Step	This note	Manuscript / SI	Code
Trap Hamiltonian, orbitals	eqs N1–N3	SI eqs S1–S4	analytic; `TrapS` (03)
Density fields $n, \nabla n, \nabla\nabla n, \tau$	–	SI eqs S16, S17	`density_fields` (03)
Exact exchange, $N = 2$	eq N4	main eq A9	`self_repulsion`, `build_wire`
Gaussian product, moments	eqs N5–N8	SI eqs S5–S7	`fact2`, `integral`, `integral2`
Laplace transform, factorization	eqs N11–N13	SI eqs S8–S10	`IEE` (structure)
x_1 and x_2 integrals, closed I_k	eqs N14–N18	SI eqs S11, S12	`integral3`
Boys connection (isotropic)	eqs N19, N20	–	–
Default Boys-type integral in t	eq N21	SI eq S13	`integrand2`, `IEE`
Failure analysis	eqs N22, N23	main Sec. 3	–
Proposed integral in $s = \ln u$	eqs N24, N25, N26	main eq 1	`IEE_robust`
Exact test integral	eq N27	SI eq S15, main eq A44	`self_repulsion`
Closed forms	–	main eqs A10, A11	`ex_exact_sheet`, `ex_exact_wire`
Comparison table, figure	Table N1, Fig. N1	main Table 4	`iee_rows`, `make_table_and_figure`

8 Where the codes are: `SCAN_1D_2D/codes/01-09`

All code is in the folder `codes/` as nine standalone single-file Python scripts. Each defines every function it uses (the aniGTO integrals are copied verbatim) and imports only third-party packages, so one command reproduces its data, e.g. `cd codes && python3 04_exact_vs_proposed.py`. Outputs go to `codes/results/` (CSV, logs, LaTeX table bodies) and `codes/figures/`. None of the scripts solves a Kohn–Sham or SCF equation (Section 1); PySCF is used only to evaluate libxc functionals on the given density.

Table N3: The nine scripts and what they produce.

Script	Computes (manuscript item)	Needs
`01_wire_benchmark.py`	wire reference quadrature of LDA/SCAN/DC-SCAN; SI Tables S1–S3, SI Figure S1	numpy, scipy
`02_sheet_benchmark.py`	sheet benchmark; main Table 2, SI Tables S5, S6, SI Figure S2	numpy, scipy
`03_anigto_wire_libxc.py`	wire with aniGTO ingredients: analytic trap orbital, exact exchange from the aniGTO ERI, density derivatives, libxc functionals; main Table 1, SI Table S4	numpy, scipy, pyscf
`04_exact_vs_proposed.py`	**this note**: default aniGTO ERI versus the $s = \ln u$ ERI, and the $N = 2$ SCE solver; main Table 4, Figure 1	numpy, scipy
`05_refit_switch_scales.py`	PBE base and refit of the activation scales; SI Table S8	numpy, scipy
`06_range_separation_q1d_lda.py`	range separation and the quasi-1D LDA (SI Section S6)	numpy, scipy
`07_q1d_gga_profiles.py`	quasi-1D LDA/GGA on seven axial profiles; SI Table S9	numpy, scipy
`08_local_hybrid_wires.py`	local hybrids and DC-SCAN on $N = 2, 4, 6$ wires (fixed trap orbitals); SI Table S10	numpy, scipy, pyscf
`09_literature_validation.py`	sheet-limit construction against Constantin and Kaplan–Wagle–Perdew; main Table 3, SI Table S7	numpy, scipy

The whole pipeline is rerun by `./refresh_all.sh`, which also rebuilds the manuscript, the SI and this note.

References

[1] S. F. Boys, Electronic wave functions. I. A general method of calculation for the stationary states of any molecular system, *Proc. R. Soc. London A* **200**, 542–554 (1950), DOI: 10.1098/rspa.1950.0036.

[2] G. Chirchir, S. Jana, S. Śmiga, An Anisotropic Gaussian Integral Package for Quasi-Dimensional Quantum-Chemical Simulations, manuscript in preparation (2026).